

Fetch API in JS

1. What is the fetch API?

- a. The **Fetch API** is a modern way to get data from a server (like an API or a file).
- b. It allows you to make network requests and retrieve data.
- c. **Example:** Getting JSON data from an api.
- d. **Syntax:** `fetch("url")`
- e. Before Fetch, developers used **XMLHttpRequest (AJAX)**, which was harder.
- f. Now Fetch is **simpler, cleaner, and preferred**.
- g. So fetch API is an alternative solution (option) for XMLHttpRequest (AJAX).

2. Key Points About the fetch API:

- a. **Promise-Based:**
 - i. Fetch uses **promises** to handle success or failure.
 - **Success** → gives you data.
 - **Failure** → gives you an error.
- b. **Returns a Response Object:**
 - i. When you call `fetch()`, it gives back a **response object**.
 - ii. From this, you can read data, status, etc.
- c. **No More AJAX:**
 - i. No need for the old **XMLHttpRequest**.
 - ii. Fetch is **easier and more flexible**.

3. How to Use the fetch API:

- a. There are two common ways to use the fetch API:
 - i. Using `.then()` and `.catch()` for promise chaining.
 - ii. Using `async` and `await` for a cleaner, synchronous-looking style.

4. Example-1: Using `.then()` and `.catch()`

```
fetch("https://api.github.com/users")
  .then((response) => {
    // Convert the response into JSON
    return response.json();
  })
  .then((data) => {
    // Use the data
    console.log(data);
  })
  .catch((error) => {
    // Handle any errors
    console.log(error);
  });
```

Explanation:

1. `fetch("url")`: Sends a request to the given URL.
2. `.then((response) => response.json())`: Converts the raw response into JSON format (which is easier to work with in JavaScript).
3. `.then((data) => { ... })`: Access the actual data and process it (e.g., log it to the console).
4. `.catch((error) => { ... })`: Handles any errors, like a network issue.

5. Example-2: Using async and await keyword

```
#!/ Create a function which will fetch the data
async function fetchData() {
  try {
    /*! Step-1: Fetch the data
      let response = await fetch("https://api.github.com/users");

    /*! Step-2: Convert the response into JSON
      let data = await response.json();

    /*! Step-3: Use the data
      console.log(data);
    } catch (error) {
      /*! Handle errors
      console.log("Error while fetching:", error);
    }
  }

  /*! Call the fetchData function
  fetchData();
```

Explanation:

1. **async** function: Allows us to use **await** for handling promises.
2. **await fetch("url")**: Waits for the fetch request to complete and returns the response.
3. **await response.json()**: Waits for the response to be converted to JSON format.
4. **try...catch**: Handles errors gracefully, if something goes wrong.